MAT627 Chapter 2.5 Programming Assignment

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1 Part 1

Composite trapezoid rule for $f(x) = x^2 e^{-x}$ on $[1, 3]$.

Table 1: Composite trapezoid-rule results on $[1, 3]$.

N	h	$T_N(f)$	$E_h = \lvert T_N(f) - I(f)\rvert$	Rate
2	1	9.493227×10^{-1}	4.369438×10^{-2}	–
4	0.5	9.821984×10^{-1}	1.081866×10^{-2}	2.013926
8	0.25	9.903203×10^{-1}	2.696735×10^{-3}	2.004236
16	0.125	9.923434×10^{-1}	6.736653×10^{-4}	2.001110
32	0.0625	9.928487×10^{-1}	1.683836×10^{-4}	2.000281

2 Part 2

Composite trapezoid rule for $f(x) = x^2 e^{-x}$ on $[0, 2]$.

Table 2: Composite trapezoid-rule results on $[0, 2]$.

N	h	$T_N(f)$	$E_h = \lvert T_N(f) - I(f)\rvert$	Rate
2	1	6.385500×10^{-1}	8.097160×10^{-3}	–
4	0.5	6.461128×10^{-1}	5.344012×10^{-4}	3.921421
8	0.25	6.466133×10^{-1}	3.386396×10^{-5}	3.980100
16	0.125	6.466450×10^{-1}	2.123833×10^{-6}	3.995009
32	0.0625	6.466470×10^{-1}	1.328545×10^{-7}	3.998751

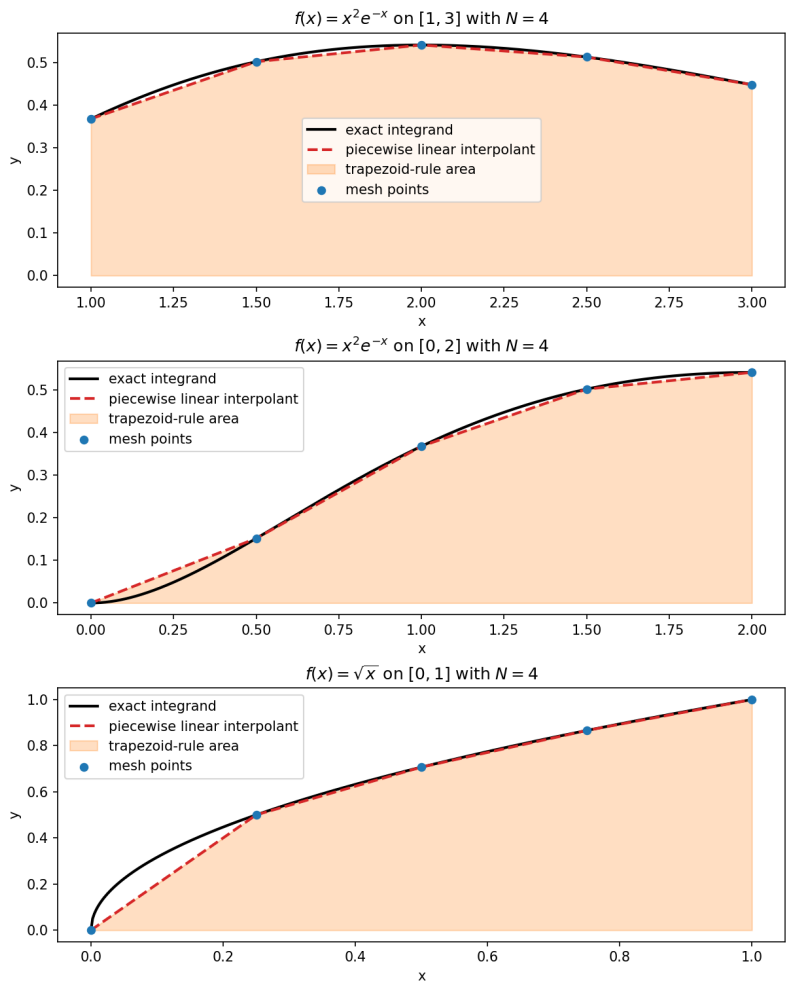

Figure 1: Representative trapezoid-rule geometry using the piecewise linear interpolant with $N = 4$.

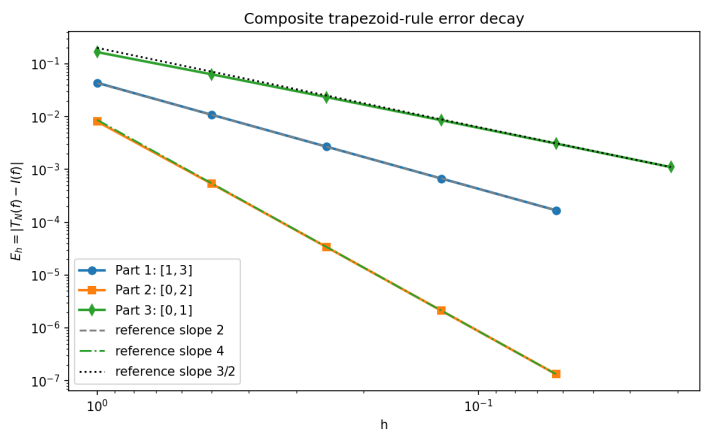

Figure 2: Error decay for the three test problems.

3 Part 3

Composite trapezoid rule for $f(x) = \sqrt{x}$ on $[0, 1]$.

Table 3: Composite trapezoid-rule results on $[0, 1]$.

N	h	$T_N(f)$	$E_h = \lvert T_N(f) - I(f)\rvert$	Rate
1	1	5.000000×10^{-1}	1.666667×10^{-1}	–
2	0.5	6.035534×10^{-1}	6.311328×10^{-2}	1.400950
4	0.25	6.432830×10^{-1}	2.338362×10^{-2}	1.432445
8	0.125	6.581302×10^{-1}	8.536445×10^{-3}	1.453791
16	0.0625	6.635812×10^{-1}	3.085470×10^{-3}	1.468145
32	0.03125	6.655589×10^{-1}	1.107730×10^{-3}	1.477883

4 Discussion

For Parts 1 and 2, the integrand is

$$f(x) = x^2 e^{-x}, \qquad f''(x) = (x^2 - 4x + 2)e^{-x},$$

so f'' is continuous and bounded on both closed intervals. Therefore Theorem 2.3 guarantees at least second-order convergence.

In Part 1, the observed rate approaches 2, which matches the theorem directly.

In Part 2, the observed rate is closer to 4. This means the $O(h^2)$ term vanished. Indeed,

$$f'(x) = (2x - x^2)e^{-x},$$

so

$$f'(0) = f'(2) = 0.$$

The leading trapezoid-rule error term is proportional to $f'(b) - f'(a)$, so on $[0, 2]$ it cancels and the next nonzero term gives fourth-order behavior.

For Part 3, the integrand is

$$f(x) = \sqrt{x}, \qquad f''(x) = -\frac{1}{4x^{3/2}},$$

which is singular at $x = 0$. Theorem 2.3 does not apply on $[0, 1]$. On the first subinterval $[0, h]$, the trapezoid rule uses the line through $(0, 0)$ and $(h, \sqrt{h})$, so

$$T_{[0,h]} = \frac{h}{2}\sqrt{h} = \frac{1}{2}h^{3/2}, \qquad \int_0^h \sqrt{x}\, dx = \frac{2}{3}h^{3/2}.$$

Hence the first cell alone contributes an error of

$$\frac{2}{3}h^{3/2} - \frac{1}{2}h^{3/2} = \frac{1}{6}h^{3/2},$$

which explains why the overall convergence rate is approximately $3/2$ rather than 2.